DS312 Final Project Guide

Data Mining and Application | 1st Semester A.Y. 2026–2027
Instructor: Nicole S. Menorias
Project window: October 12 – December 5, 2026

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1. Project Overview

In this project your group will act as a small data science consultancy. You will find a real dataset, define a real business problem, clean the data, explore it, and apply data mining (and optionally machine learning) techniques to produce findings someone could act on. This will serve as your final exam.

The project follows the full data mining process taught in this course:
Business problem → Data → Preprocessing → Exploration → Modeling → Interpretation → Communication

Final deliverables (as a group)

1. **Jupyter Notebook:** one master notebook documenting all four phases, with explanations of *why* you made each choice.
2. **Poster:** a visual summary of the whole project.
3. **PowerPoint presentation:** delivered to the class during Final Project Consultation/Presentation.

Every Phase Deliverables

1. **Individual Narrative Report (individual):** what *you specifically* contributed in that phase.
2. **Peer Evaluation (individual):** your confidential rating of each teammate for that phase.
3. **Phase Notebook (group)**: only the leader will pass this

2. Project Timeline

Phase	Focus	Dates	Group deadline
1	Dataset and Business Problem	Oct 12 – Oct 24	Sun, Oct 25
2	Data Preprocessing	Oct 26 – Nov 7	Sun, Nov 8
3	Exploratory Data Analysis (EDA)	Nov 9 – Nov 14	Sun, Nov 15
4	Data Mining / Machine Learning	Nov 16 – Nov 28	Sun, Nov 29
5: Final	Poster, PPT, final notebook, presentation	Nov 30 – Dec 5	Presentation per class schedule

Note: Individual narrative reports and peer evaluations are due **1 day after each group deadline**.

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3. Phase 1: Dataset Selection and Business Problem

Dates: Oct 12 – Oct 24

Objective

Find a suitable dataset and define a clear, realistic business (or social) problem that data mining can help solve.

Tasks

1. **Brainstorm** 2–3 candidate problems and datasets as a team.
2. **Evaluate each dataset** using the criteria below and pick one.
3. **Write the business problem** using the canvas below.
4. **Describe the data**: source, size, columns, data types, collection method, date range, license.
5. **Propose a data mining plan**: which technique families could answer your questions, and why.

6. **Consult the instructor** before finalizing (a short approval check, see section 10).

Dataset selection criteria

Criterion	Minimum expectation
Size	At least ~5,000 rows (larger is fine)
Features	At least ~10 columns with a mix of numeric and categorical
Messiness	Has realistic issues (missing values, inconsistent formats, duplicates, outliers). A perfectly clean dataset is a red flag, not a bonus.
Relevance	Connects to a real decision, organization, or community

Avoid overused teaching datasets (Titanic, Iris, Boston Housing, Wine, MNIST) as your main dataset.

Dataset Quality and Complexity (scored)

The quality, richness and difficulty of your dataset are graded: the more complex the data you take on, the more points you can earn, as long as your group handles it well. A simple, clean dataset caps out at the lower tier however well it is analyzed.

Tier	What the dataset looks like	Dataset score (of 8)
Basic	Single table, mostly clean, few data types, a handful of simple issues (some missing values, duplicates)	3–4
Moderate	10,000+ rows, 15+ columns, mixed types (numeric, categorical, dates), several kinds of issues (missing values, inconsistent categories, outliers, wrong types)	5–6
Advanced	Multiple tables or sources to integrate, or text, date-time or geospatial fields; heavy or patterned missingness; inconsistent coding across sources; large volume or class imbalance; needs domain knowledge to interpret	7–8

The score considers volume and variety, number of sources, real-world messiness, domain depth, relevance to your problem, and how well the data's origin is documented.

Rules

- **Difficulty must be genuine.** Do not corrupt, shuffle or inject errors into a clean dataset to make it look harder. That is academic dishonesty.
- **Complexity earns points only when handled well.** An advanced dataset that is poorly cleaned or poorly explained scores lower than a moderate dataset handled thoroughly.
- **Justify your tier** in a short *Dataset Complexity Statement* in your Phase 1 notebook section: what makes this data hard, with evidence (number of sources, missing-value counts, data types, row volume).
- **Check feasibility with the instructor.** A dataset too large or too messy to finish in two weeks is a risk to your group.

Where to look

- Kaggle Datasets and UCI Machine Learning Repository
- Philippine Statistics Authority (PSA) OpenSTAT and the Philippine Open Data portal (data.gov.ph)
- World Bank Open Data
- Humanitarian Data Exchange (HDX)
- Agency or LGU open data (health, transport, education, environment)

Business Problem Canvas (include in your notebook)

Item	Your answer
Context / organization	Who has this problem?
Problem statement	One or two sentences.
Stakeholders	Who will use the results?
Business questions	3–5 specific questions the data should answer
Success criteria	How will we know the analysis is useful?
Task type: Data mining or Machine Learning or Both	Clustering, association rules, anomaly detection, classification, regression, etc.
Risks and limitations	Bias, missing populations, privacy, data age

Deliverables (group notebook, Section 1)

- Problem statement and canvas
- Dataset description and data dictionary
- Dataset evaluation (why this one, why not the alternatives)
- Preliminary data mining plan
- Initial load of the data with .head(), .info(), .shape

Checklist

- ☐ The problem is specific and answerable with the data
- ☐ Dataset meets the criteria, with the source cited
- ☐ Data mining plan names at least two candidate techniques with reasons
- ☐ Instructor has approved the dataset and problem

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4. Phase 2: Data Preprocessing

Dates: Oct 26 – Nov 7

Objective

Transform raw data into a clean, analysis-ready dataset, and document every decision so another person can reproduce it.

Tasks

1. **Data profiling:** data types, missing-value counts, duplicates, unique values, ranges.
2. **Data cleaning:** handle missing values (drop, impute, or flag, and justify), remove duplicates, fix inconsistent categories and formats, correct data types.
3. **Outlier treatment:** detect (IQR, z-score, domain rules) and decide what to do. Do not delete outliers blindly; some are the finding.
4. **Data integration** (if applicable): merge multiple sources and check join keys and row counts before and after.
5. **Transformation:** encoding categorical variables, scaling or normalization, date and time features, binning, log transforms where needed.
6. **Feature engineering:** create at least a few meaningful new features tied to your business questions.
7. **Data reduction** (if needed): drop irrelevant or redundant columns, sampling.
8. **Save outputs:** write the cleaned dataset to /data/processed and keep the raw file untouched.

Cleaning Depth and Difficulty (scored)

How deep and how difficult your data cleaning is will be graded, not just whether the data ends up clean. The harder the problems your group finds and solves properly, the more points you earn.

Level	Typical work	Cleaning score (of 10)
Basic	Drop rows with missing values or fill with the mean or median, remove duplicates, rename columns, simple type conversions	3–4
Intermediate	Standardize inconsistent categories, parse dates and times, detect outliers (IQR, z-score) and decide with reasoning, group-based or justified imputation, encoding, scaling, derived features	5–7
Advanced	Integrate multiple sources (key reconciliation, join validation), fuzzy matching or entity resolution, model-based imputation (KNN, iterative), text cleaning, handling patterned missingness, domain-rule validation, a reusable cleaning pipeline (functions or a scikit-learn pipeline), automated data quality checks (assertions)	8–10

The score weighs five things:

1. **Number and variety** of data problems found and addressed
2. **Difficulty and sophistication** of the methods used
3. **Justification:** each step tied to evidence of a problem, with alternatives compared
4. **Impact:** proof of the before-and-after difference
5. **Reproducibility:** the notebook runs top to bottom

Data Quality Report (required). Include a before-and-after table covering rows, columns, missing values (%), duplicates, outliers flagged, inconsistent categories fixed, and type corrections.
Rule: Complexity for show earns nothing. Every cleaning step must tie to a documented problem in your data, and every advanced method must be justified over a simpler one.

Deliverables (group notebook, Section 2)

- Before-and-after summary of the data (shape, missing values, dtypes)
- A **preprocessing log**: a table listing each issue found, the action taken, and the reason
- Reproducible code (the notebook runs top to bottom without errors)
- The cleaned dataset file

Preprocessing log example

Issue	Columns	Action	Justification
12% missing values	income	Median imputation by region	Skewed distribution; median is robust
Inconsistent labels	city	Standardized spelling and case	"CDO", "Cagayan de Oro" and "cdo " referred to the same city

Checklist

- ☐ Every cleaning decision is justified, not just performed
- ☐ Raw data is preserved
- ☐ Notebook runs end to end
- ☐ No information from the future or from the target is leaked into features (check this early)

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5. Phase 3: Exploratory Data Analysis (EDA)
Dates: Nov 9 – Nov 14

Objective
Understand the patterns, relationships, and anomalies in your cleaned data, and use them to refine your modeling plan.

Tasks

1. **Univariate analysis:** distributions, summary statistics, frequency counts.
2. **Bivariate and multivariate analysis:** correlations, group comparisons, cross-tabulations, pairwise relationships.
3. **Visualization:** at least **8 purposeful visualizations** with titles, labeled axes, and a one-to-two-sentence interpretation under each.
4. **Hypothesis testing:** state at least **two hypotheses** from your business questions and test them statistically (for example t-test, chi-square, correlation test). Report the test, the p-value, and a plain-language conclusion.
5. **Pattern spotting:** segments, trends, seasonality, suspicious values, class imbalance.
6. **Answer or refine your business questions** from Phase 1 using what you found.
7. **Modeling implications:** which features look useful, which techniques look promising, what problems (imbalance, multicollinearity) must be handled.

Deliverables (group notebook, Section 3)

- Organized EDA with narrative text between code cells
- Visualizations with interpretations
- Hypothesis tests with conclusions
- A closing **"Key Insights and Modeling Plan"** section (5–8 bullet insights and the chosen techniques)

Checklist

- ☐ Each chart answers a question; none is decoration
- ☐ Insights are stated in words, not just shown
- ☐ Correlation is not presented as causation
- ☐ Findings link back to the business problem

6. Phase 4: Data Mining / Machine Learning
Dates: Nov 16 – Nov 28

Objective
Apply appropriate data mining techniques to answer your business questions, evaluate them honestly, and interpret the results for the stakeholders.

Requirements

- Apply **at least two techniques from different families** covered in this course (for example, clustering plus classification, or association rules plus anomaly detection).
- Justify **why** each technique fits your problem.
- Establish a **baseline** (for example majority class, mean prediction, or a simple model) and compare your model to it.
- Evaluate using appropriate metrics and validation, and interpret the results in business terms.

Ensemble methods, deep learning, and other ML approaches are welcome **in addition to** the above, provided you can explain how they work and why you used them.

Tasks

1. **Prepare modeling data:** feature selection, train/test split (or cross-validation), apply scaling or encoding *inside* the pipeline to prevent leakage.
2. **Build the baseline.**

3. **Build and tune models:** try at least two settings or algorithms per technique; record what you tried.
4. **Evaluate:** metrics, comparison tables, visual diagnostics.
5. **Interpret:** feature importance or cluster profiles; what the result means for the stakeholder.
6. **Reflect on limitations:** bias, data quality, generalizability, ethical implications of acting on the results.
7. **Recommendations:** 3–5 concrete, actionable recommendations tied to your findings.

Deliverables (group notebook, Section 4)

- Modeling code, comparison tables, and visualizations
- A results interpretation section written for a non-technical reader
- Limitations and ethical considerations
- Conclusions and recommendations

Checklist

- ☐ Techniques are justified by the problem, not by trend
- ☐ Baseline comparison is included
- ☐ Test data was never used to make modeling decisions
- ☐ Results are interpreted in business terms
- ☐ Limitations are honest

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7. Final Deliverables

7.1 Master Jupyter Notebook

Suggested structure:

1. Title, team
2. Executive summary
3. **Phase 1:** Business problem and dataset
4. **Phase 2:** Preprocessing (with preprocessing log)
5. **Phase 3:** EDA
6. **Phase 4:** Data mining / ML
7. Conclusions, recommendations, limitations
8. References (cite the dataset and any sources)

Notebook standards (aligned with the Laboratory Exercises rubric): clear headings and sections, commented code, explanations of steps and results, no errors, and it should run top to bottom. Use Markdown cells to explain *why*, not just *what*.

7.2 Poster

A visual, engaging summary readable in under two minutes. Required sections:

- Project title and team members
- Executive summary
- Business problem and dataset
- Methodology (preprocessing steps and EDA highlights)
- Data mining / ML results (key charts and metrics)
- Conclusions and recommendations

Design tips: one main message, large readable figures, minimal text, consistent colors, readable from a few feet away.

7.3 PowerPoint Presentation

Suggested outline (about 10–12 slides):

1. Title
2. Problem and stakeholders
3. Dataset overview
4. Preprocessing: key decisions

5. EDA: top 2–3 insights
6. Technique selection and why
7. Results: technique 1
8. Results: technique 2
9. Comparison and interpretation
10. Recommendations
11. Limitations and ethics
12. Conclusion and Q&A

Every member should speak. Presentation criteria follow the course's **Reporting rubric**: presence, language skills, organization, mastery of the subject matter, visual aids, and overall impression. Be ready to answer questions about any part of the project, not only your own.

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8. Individual Requirements Per Phase

Every phase, each student submits two individual documents.

8.1 Individual Narrative Report

Length: 300–500 words. **Submitted individually** through USTeP, 3 days after each group deadline.

Purpose: to show what *you* did, not what the group did. Follow the outline below:

1. My specific contributions. List the tasks you personally completed. Be concrete (for example, "wrote the imputation code for income and age," not "worked on cleaning").

2. Evidence. Point to where your work is: notebook section or cell numbers, GitHub commits or files, charts you created.

3. Decisions I made or influenced, and why. What alternatives did you consider?

4. Challenges and how I handled them.

5. What I learned. Link to a concept from lecture or lab.

6. Time spent and my estimated share of the group's work this phase (%).

How it is graded (20 points per report):

Criterion	Points
Specificity and clarity of contribution	8
Evidence matches the group's notebook and repository	5
Reasoning and reflection (decisions, challenges, learning)	5
Writing quality and timeliness	2

Reports that contradict the repository history or the peer evaluations will be followed up with the student.

8.2 Peer Evaluation

Please access the Peer Evaluation Portal and select the corresponding project phase to complete your assessment.

[WAITS LANG GUYSES, LINK TO BE FOLLOWED]